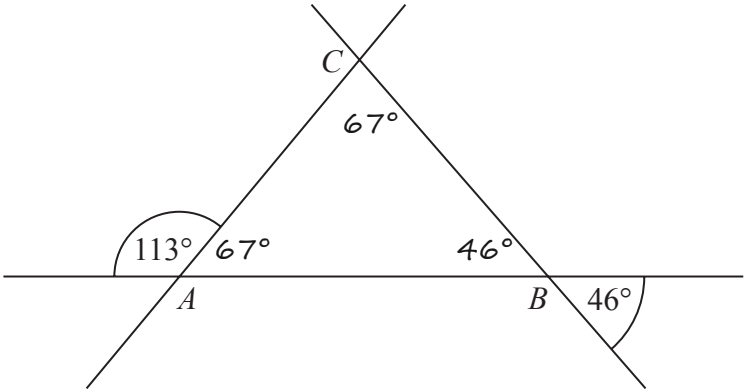
12 Here is triangle *ABC* with each of its sides extended.



Show that triangle *ABC* is isosceles.
Give a reason for each stage of your working.

Angle ABC = 46°.

Vertically opposite angles are equal.

1 mark
Find another angle

Angle BAC = 180° – 113° = 67°.

Angles on a straight line add to 180°.

1 mark
All appropriate reasons

Angle ACB = 180° – 67° – 46° = 67°.

Angles in a triangle add to 180°.

1 mark
Angle C = 67°

BAC = ACB, so ABC has two equal angles and is isosceles.

1 mark
Equal angles; isosceles

(Total for Question 12 is 4 marks)